V K - spațiu vectorial $\dim_K V = n$

$v_1, \dots v_n \in V$ vectori liniar independenți $\Leftrightarrow$ bază.

$Sp\{v_1, \dots v_n\} = V$

"$\subseteq$" merge

"$\supseteq$" Fie $v \in V$ $v_1, v_2, \dots v_n$

$\underbrace{v_1, \dots, v_n, v}_{n+1}$ liniar independenți ? $/\Rightarrow$

$\dim_K V = n$

$\Rightarrow v_1, \dots v_n, v$ liniar dependenți

$\exists \lambda_1 \dots \lambda_n, \lambda \in K$ nu toți nuli a.î.

$\lambda_1 v_1 + \dots + \lambda_n v_n + \lambda v = 0$

$\lambda = 0 \Rightarrow \lambda_1 v_1 + \dots + \lambda_n v_n = 0$

$v_1, \dots v_n$ liniar indep. $/\Rightarrow$

$\Rightarrow \lambda_1 = \dots = \lambda_n = 0$, fals

$\lambda \neq 0$ $\lambda_1 v_1 + \dots + \lambda_n v_n + \lambda v = 0 \; / \cdot \lambda^{-1}$

$\Rightarrow v_2 = \lambda_1 \lambda^T v_1 + \dots + \lambda_n \lambda^T v_n$

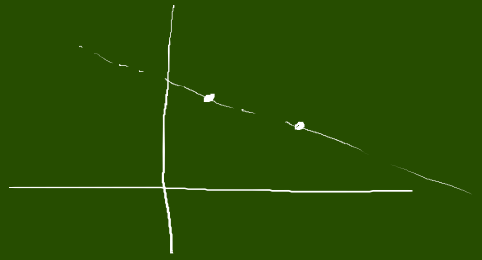
$\Rightarrow v \in Sp\{v_1, \dots, v_n\}$

$\Rightarrow V = Sp\{v_1, \dots, v_n\} \Rightarrow \{v_1, \dots, v_n\}$ bază

Exemple 1) $(x_1, y_1), (x_2, y_2)$ puncte distincte

$\Rightarrow$ există o unică dreaptă care trece

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$$



$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

2) $(x_1, y_1), (x_2, y_2), (x_3, y_3) \Rightarrow$ parabolă



3) $(x_1, y_1) \dots (x_n, y_n)$

$x_i \neq x_j$ $i \neq j$

$\Rightarrow \exists!\ P \in \mathbb{R}_{n-1}[X]$

$P(x_i) = y_i \quad \forall i = \overline{1, n}$

(interpolare Lagrange)

Fie $P_1(x) = \dfrac{(x-x_2)(x-x_3)\cdots(x-x_{n+1})}{(x_1-x_2)(x_1-x_3)\cdots(x_1-x_{n+1})} \in \mathbb{R}[x]$

grad $P_1 = n-1$ $\qquad P_1(x_1) = 1$

$P_1(x_i) = 0, \ i \neq 1$

$P_2(x) = \dfrac{(x-x_1)(x-x_3)\cdots(x-x_{n+1})}{(x_2-x_1)(x_2-x_3)\cdots(x_2-x_{n+1})} \in \mathbb{R}_{n-1}[x]$

$P_2(x_2) = 1 \qquad P_2(x_i) = 0, \ i \neq 2$

$\cdots\cdots\cdots$

$P_i(x) = \dfrac{\prod_{j \neq i}(x-x_j)}{\prod_{j \neq i}(x_i-x_j)} \in \mathbb{R}_{n-1}[x]$

$P_i(x_k) = \delta_{ik} = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$

Kronecker

astfel obținem $P_1, \ldots, P_{n+1} \in \mathbb{R}_{n-1}[x]$

$P_1 \dots P_n$ bază : $\dim \mathbb{R}_{n-1}[X] = n$

$\mathbb{R}(\{1, X, \dots, X^{n-1}\})$

$\Rightarrow$ e suf să verificăm $P_1 \dots P_n$ liniar indep.

fie $\lambda_1 \dots \lambda_n \in \mathbb{R}$ a.î. $\lambda_1 P_1 + \dots + \lambda_n P_n = 0$

$\lambda_1 P_1(X) + \dots + \lambda_n P_n(X) = 0$

$X = x_1 \quad \lambda_1 \underbrace{P_1(x_1)}_{1} + \lambda_2 \underbrace{P_2(x_1)}_{0} + \dots + \lambda_n \underbrace{P_n(x_1)}_{0} = 0$

$\Rightarrow \lambda_1 = 0$

analog $\lambda_2 = \dots = \lambda_n = 0$

$\Rightarrow$ $P_1 \dots P_n$ bază în $\mathbb{R}_{n-1}[X]$

$\exists!$ $P \in \mathbb{R}_{n-1}[X]$ $P(x_i) = y_i$ $i = \overline{1,n}$

$P = \alpha_1 P_1 + \dots + \alpha_n P_n$ $\alpha_i \in \mathbb{R}$ $i = \overline{1,n}$

$P(x_i) = y_i$

$\alpha_1 \underbrace{P_1(x_i)}_{0} + \dots + \alpha_i \underbrace{P_i(x_i)}_{1} + \dots + \alpha_n \underbrace{P_n(x_i)}_{0} = y_i$

$\Rightarrow \alpha_i = y_i$

deci $P(x) = \sum_{i=1}^{n} y_i \cdot P_i(x)$ este polinomul căutat.

unicitatea: dacă $\exists\ P, Q$ cu proprietatea

$P(x_i) = Q(x_i) = y_i \quad i = \overline{1,n}$

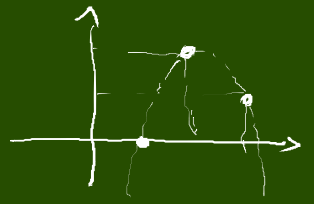
$P = \sum \alpha_i P_i \qquad Q = \sum \beta_i P_i$

$P(x_i) = \alpha_i \qquad Q(x_i) = \beta_i$

y_i

$\Rightarrow P = Q$

tema: $(1,0), (2,2), (3,1)$



$P(x) = 0 \cdot P_1(x) + 2 P_2(x) + 1 \cdot P_3(x)$

$P_1(x) = \dfrac{(x-2)(x-3)}{(1-2)(1-3)}$

$P_2(x) = \dfrac{(x-1)(x-3)}{(2-1)(2-3)}$

$P_3(x) = \dfrac{(x-1)(x-2)}{(3-1)(3-2)}$

$P(X) = \dots$

$\dim_K V = n$

$v_1 \dots v_k$ liniar independenți

$k = n$ stop bază

$k < n \Rightarrow \exists\, v \in V,\ v \notin Sp\{v_1 \dots v_k\}$

$\Rightarrow v_1 \dots v_k, \underbrace{v}_{\text{notăm } v_{k+1}}$ liniar indep.

$k+1 = n$ stop bază

$k+1 < n \Rightarrow \exists\, v' \in V,\ v' \notin Sp\{v_1 \dots v_{k+1}\}$

etc

Continuăm până când obținem

n vectori $v_1 \dots v_n$ liniar independenți $\Big| \Rightarrow \dim_K V = n$

$\Rightarrow$ bază.

Exemplu. $V = \mathbb{R}_2[X]$ $\dim_{\mathbb{R}} V = 3$

$P_1(X) = 1 - X$, $P_2(X) = 1 + X^2$

Teoremă $f : V \to V$ izomorfism

$u_1, \dots u_n \in V \Rightarrow$

a) $u_1, \dots u_n$ lin. indep $\Leftrightarrow f(u_1) \dots f(u_n)$ lin. indep

b) $u_1, \dots u_n$ sist. gen. $\Leftrightarrow f(u_1), \dots f(u_n)$ sist. gen.

c) $u_1, \dots u_n$ bază $\Leftrightarrow f(u_1), \dots f(u_n)$ bază.

$$f : \mathbb{R}_2[X] \to \mathbb{R}^3$$

$$1 \to e_1$$

$$X \to e_2$$

$$X^2 \to e_3$$

$$f(a_0 + a_1 X + a_2 X^2) = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$$

$P_1(x) = 1 - x$, $P_2(x) = 1 + x^2$ lin. indep.

$\Leftrightarrow \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ lin. indep $\Leftrightarrow$

$\operatorname{rang} \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & 1 \end{pmatrix} = 2$ ✓

Completer $\underbrace{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{v_2}$ to a base :

$\det \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \neq 0$ For $v_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\Rightarrow v_1, v_2, v_3$ base of $\mathbb{R}^3$

$v_3 = f(P_3)$ $P_3(x) = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2 = 1$

$\Rightarrow$ base of $\mathbb{R}_2[x]$ $P_1(x) = 1 - x$, $P_2(x) = 1 + x^2$

$P_3(x) = 1$

Exemplu, $\mathbb{R}_2[X]$

$Q_1(X) = 1 - X + X^2 \longrightarrow w_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

$Q_2(X) = 2 + 3X \longrightarrow w_2 = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$

$Q_3(X) = -1 + X^2 \longrightarrow w_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

$Q_4(X) = X + 3X^2 \longrightarrow w_4 = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$

Teorema: $Sp\{Q_1, Q_2, Q_3, Q_4\} = \mathbb{R}_2[X]$

$\dim_{\mathbb{R}} \mathbb{R}_2[X] = 3$

$\lambda_1 Q_1 + \lambda_2 Q_2 + \lambda_3 Q_3 = Q_4$

$$\Rightarrow \begin{cases} \lambda_1 + 2\lambda_2 - \lambda_3 = 0 & 1 \\ -\lambda_1 + 3\lambda_2 = 1 & X \\ \lambda_1 + \lambda_3 = 3 & X^2 \end{cases}$$

$$\operatorname{rang} \underbrace{\begin{pmatrix} 1 & 2 & -1 & 0 \\ -1 & 3 & 0 & 1 \\ 1 & 0 & 1 & 3 \end{pmatrix}}_{w_1 \; w_2 \; w_3 \; w_4} = 3$$

$\det(\mathrm{id}_1 / \mathrm{id}_2 / w_3) \neq 0 \Rightarrow w_1, w_2, w_3$ base de $\mathbb{R}_3$

$\Rightarrow \underline{Q_1}, \underline{Q_2}, \underline{Q_3}$ base de $\mathbb{R}_2[x]$

$U \subseteq V \qquad \dim_k V < \infty$

$\dim_k U \leq \dim_k V$

$\dim_k U = \dim_k V \iff U = V$

Exemples $\qquad V = \mathbb{R}[x] \quad \{1, x, x^2, \dots, x^n, \dots\}$

$\dim_{\mathbb{R}} V = \infty$

$U = \{P \in \mathbb{R}[x] \mid P(x) = P(-x)\}$

polinom par

$P(x) = a_0 + a_2 x^2 + \dots + a_{2n} x^{2n}$

$U = \{1, x^2, x^4, \dots, x^{2n}, \dots\}$

$\dim_{\mathbb{R}} U = \infty$

$\Rightarrow U = V$? NU ex. $P(x) = X \notin U$

$\dim_k U = \dim_k V = \infty$ dar $U \neq V$

$A \in \mathcal{M}_{m,n}(k) \Rightarrow f : k^n \longrightarrow k^m$

$f(x) = Ax$

$\ker(f) = \ker(A) \qquad \operatorname{Im}(f) = \operatorname{Im}(A)$

$\dim_k \ker(f) + \underbrace{\dim_k \operatorname{Im}(f)}_{\operatorname{rang}(A)} = n = \dim_k k^n$

Exemplu: $A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 2 \\ -1 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}$

$\ker(A) \subseteq \mathbb{R}^3 \qquad \dim \operatorname{Im}(A) = \operatorname{rang} A$

$\operatorname{Im}(A) \subseteq \mathbb{R}^4$

$\dim \ker(A) = 3 - \operatorname{rang}(A)$.

$$\ker(A) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$(4,3)\ (3,1)\ \ (4,1)$

$$\begin{cases} x - z = 0 \\ 2y + 2z = 0 \\ -x + y + 2z = 0 \\ 2x + 3y + z = 0 \end{cases} \Rightarrow \begin{array}{l} x = z \\ y = -z \\ -z - z + 2z = 0,\ 0 = 0 \\ 2z - 3z + z = 0,\ 0 = 0 \end{array}$$

$$\ker(A) = \left\{ \begin{pmatrix} z \\ -z \\ z \end{pmatrix} \;\middle|\; z \in \mathbb{R} \right\} = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow \left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\} \text{ base di } \ker(A)$$

$$\Rightarrow \dim_{\mathbb{R}} \ker(A) = 1$$

$$\Rightarrow \operatorname{rang}(A) = 3 - 1 = 2 = \dim_{\mathbb{R}} \mathrm{Im}(A)$$

$$\mathrm{Im}(A) = \mathrm{Sp}\left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 0 \\ 2 \\ 1 \\ 3 \end{pmatrix}}_{v_2}, \underbrace{\begin{pmatrix} -1 \\ 2 \\ 2 \\ 1 \end{pmatrix}}_{v_3} \right\}$$

$$\operatorname{rang}(A) = 2 = \operatorname{rang}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 2 \\ -1 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}$$

Obs : $v_2 = v_1 + v_3$

$\mathrm{Im}(A) = \mathrm{Sp}\{v_1, v_3\}$

$\Rightarrow \{v_1, v_3\}$ baza

$A = \{a_1, \dots a_n\}$ multime finita

$\mathrm{card}\, A = n$

$B \subseteq A \Rightarrow \mathrm{card}(B) \leq n = \mathrm{card}(A)$

$\mathrm{card}(B) = \mathrm{card}(A) \Rightarrow B = A$

$C_n^0 + C_n^1 + C_n^2 + C_n^3 + \dots + C_n^n = 2^n$

$2^n = \mathrm{card}\, \mathcal{P}(A)$

$\mathcal{P}(A) = \{B \mid B \subseteq A\}$

A infinita

$B \subseteq A$ B infinita $\not\Rightarrow B = A$?

nu neaparat

$A = \mathbb{N}$ $\quad B = \{2n \mid n \in \mathbb{N}\}$

$B \subseteq A$ dar $B \neq A$

A mulțime finită

$B, C \subseteq A$ $\quad B_1, \dots B_n \subseteq A$

$\text{card}(B \cup C) = \text{card}\, B + \text{card}\, C - \text{card}(B \cap C)$

(inducție.

$$\text{card}(B_1 \cup \dots \cup B_n) = \sum_{i=1}^{n} \text{card}(B_i) -$$

$$- \sum_{1 \le i < j \le n} \text{card}(B_i \cap B_j) + \sum_{i<j<k} \text{card}(B_i \cap B_j \cap B_k)$$

$$- \dots + (-1)^{n-1} \text{card}(B_1 \cap \dots \cap B_n)$$

$U \cap W$ bază B

completez $B \cup B'$ bază în U

completez $B \cup B''$ bază în W

$\Rightarrow$ $\mathcal{B} \cup \mathcal{B}' \cup \mathcal{B}''$ bază în $U+W$

- lin. indep.
- sist. gen.

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exemplu. $V = \mathbb{R}^3$

$U = Sp\{\underbrace{\begin{pmatrix}1\\0\\2\end{pmatrix}}_{u_1}, \underbrace{\begin{pmatrix}1\\2\\2\end{pmatrix}}_{u_2}\}$ plan $\quad U+W,\ U\cap W$

$W = Sp\{\underbrace{\begin{pmatrix}1\\1\\0\end{pmatrix}}_{w_1}, \underbrace{\begin{pmatrix}1\\0\\-1\end{pmatrix}}_{w_2}\}$ plan

$\{u_1, u_2\}$ bază în U, $\dim_{\mathbb{R}} U = 2$

$\{w_1, w_2\}$ bază în W, $\dim_{\mathbb{R}} W = 2$

$U \cap W$ < dr. $\subset \mathbb{R}^3$; plan

$U+W$ < $\mathbb{R}^3$; plan

ec. plan care pt U $\quad \pi_U$

var. 1 $\quad (1,0,2) \in \pi_U$ $\quad Ax+By+Cz+D=0$

$(1,2,2) \in \pi_U$ $\quad \Rightarrow A, B, C, D = \dots$

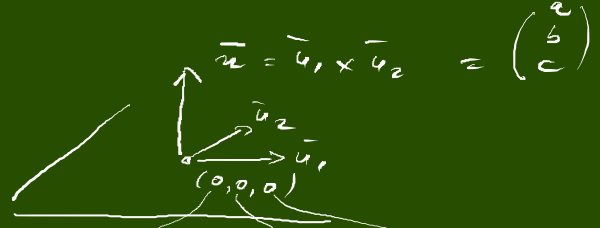
$(0,0,0) \in \pi_U$

var. 2

$$\begin{vmatrix} x & y & z & 1 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 0$$

$\Rightarrow \dots$ ec. pl. π_U

var. 3



$$a(x-0) + b(y-0) + c(z-0) = 0$$

Metoda lui Gauss pt. determinarea unei baze și dim în $U \cap V$, $U + V$

$$\left(\begin{array}{cc|c} \textcircled{1} & 0 & 2 & u_1 \\ 1 & 2 & 2 & u_2 \\ 1 & 1 & 0 & w_1 \\ 1 & 0 & -1 & w_2 \end{array}\right) \xrightarrow[L_4 - L_1]{\substack{L_2 - L_1 \\ L_3 - L_1}} \left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & \textcircled{2} & 0 & u_2 - u_1 \\ 0 & 1 & -2 & w_1 - u_1 \\ 0 & 0 & -3 & w_2 - u_1 \end{array}\right)$$

$$\xrightarrow{1/2\, L_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & \textcircled{1} & 0 & (u_2 - u_1)/2 \\ 0 & 1 & -2 & w_1 - u_1 \\ 0 & 0 & -3 & w_2 - u_1 \end{array}\right) \longrightarrow$$

$$\xrightarrow{L_3 - L_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & 1 & 0 & (u_2 - u_1)/2 \\ 0 & 0 & -2 & w_1 - \frac{u_1}{2} - \frac{u_2}{2} \\ 0 & 0 & -3 & w_2 - u_1 \end{array}\right)$$

$$\xrightarrow{L_4 - \frac{3}{2} L_3} \left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & 1 & 0 & (u_2 - u_1)/2 \\ 0 & 0 & -2 & w_1 - u_1/2 - u_2/2 \\ 0 & 0 & 0 & w_2 - u_1 - \frac{3}{2} w_1 + \frac{3}{4} u_1 + \frac{3}{4} u_2 \end{array}\right)$$

$$0 = w_2 - 3/2\, w_1 - u_1/4 + 3/4\, u_2$$

$\dim_{\mathbb{R}}(U \cap \langle x \rangle) = 1$

$\dim_{\mathbb{R}}(U + \langle x \rangle) = 3$

$$\underbrace{w_2 - 3/2\, w_1}_{W} = \underbrace{u_1/4 - 3/4\, u_2}_{U}$$

$W \cap U$

$\Rightarrow$ basis in $U \cap W$ $\quad \{ w_2 - 3/2\, w_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} - 3/2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$$= \begin{pmatrix} -1/2 \\ -3/2 \\ -1 \end{pmatrix} \}$$

basis in $U + W$ $\quad \{ u_1, (u_2 - u_1)/2, w_1 - u_1/2 - u_2/2 \}$

$U + \langle x \rangle \subseteq \mathbb{R}^3 \quad \Rightarrow \quad U + \langle x \rangle = \mathbb{R}^3$